Starhash: A Compact Analytical Celestial Seal

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Abstract

Starhash is a small Python library that turns an observer's latitude, longitude, elevation, and UTC time into a monochrome SVG view of the local sky and a separate compact digest. It uses low-order secular elements for five planets, selected terms from the ELP-2000 lunar expansion, a 457-star catalogue with a simple precession correction, topocentric parallax, atmospheric refraction, and zenith-centred stereographic projection. The implementation has no runtime dependencies. This paper describes the equations and the rendering format, gives one worked example, and reports consistency checks. The inverse routine is a local fit to labelled points generated by the same model; it is not a blind decoder or a precision ephemeris.

1 Introduction

Starhash is an astronomy script with a narrow purpose: produce a repeatable celestial drawing and a short digest from a stated place and time. The drawing is intended to be read as an object in its own right. It contains geometry and planetary symbols, but no printed name or digest.

The astronomical ingredients are established ones. The implementation combines them in a small, self-contained program rather than attempting to replace a general-purpose ephemeris. The planetary elements follow the style of compact secular models discussed by Simon et al. [5], Bretagnon and Francou [1]; the lunar terms are selected from ELP-2000 [3]; and the rotation and obliquity expressions are reduced forms of standard IAU formulae [2, 4]. The arrangement of these ingredients into the Starhash seal is the contribution here.

2 Model

2.1 Time and rotation

Dates are converted to Julian Day using the proleptic Gregorian calendar. The implementation uses

$$T = \frac{JD - 2451545.0}{36525} \tag{1}$$

for the number of Julian centuries from J2000.0. UTC is used as the time argument for this compact model; no ΔT, UT1 correction, or Earth-orientation table is applied.

The rotation angle is the IAU Earth Rotation Angle expression

$$\theta = 2\pi\big(0.779057273264 + 1.00273781191135\,d\big) \pmod{2\pi}. \tag{2}$$

where $d = JD - 2451545.0$. The local angle used by the horizontal-coordinate transform is $\theta + \lambda$, with east longitude λ. It is an ERA-based rotation proxy, not a complete apparent sidereal time. The mean obliquity is evaluated with the truncated polynomial

$$\varepsilon = \big(84381.406 - 46.836769T - 0.0001831T^2 + 0.00200340T^3\big)''. \tag{3}$$

2.2 Planets and Moon

For Mercury, Venus, Mars, Jupiter, and Saturn, a small set of secular orbital elements gives the semimajor axis, eccentricity, inclination, node, longitude of perihelion, and mean longitude. The eccentric anomaly E is obtained from Kepler's equation,

$$M = E - e\sin E, \tag{4}$$

and the resulting heliocentric vector is subtracted from the corresponding Earth vector. The ecliptic vector is then rotated by ε to obtain right ascension and declination. These are low-order approximations, not a full VSOP87 evaluation.

The lunar position uses selected periodic terms from the ELP-2000 expansion. In the notation of that expansion,

$$\ell = \ell_0 + 10^{-6}\sum L_i, \tag{5}$$

$$b = 10^{-6}\sum B_i, \tag{6}$$

$$\Delta = 385000.56 + 10^{-3}\sum R_i \text{ km}. \tag{7}$$

The unit factors are part of the model: longitude and latitude coefficients are in microdegrees, while distance coefficients are in 0.001 km. The lunar phase and bright-limb angle are derived from the solar and lunar directions.

For a finite-distance body, the observer's position is reduced on a WGS84 ellipsoid. The correction uses the horizontal parallax $\pi = \arcsin(a/\Delta)$ and the local hour

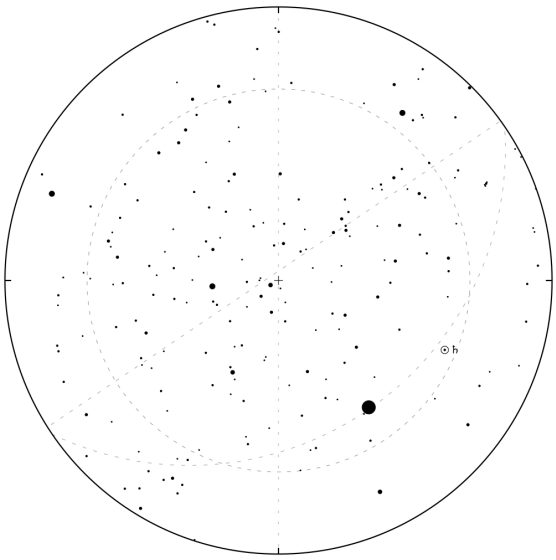


Figure 1: Label-free Starhash seal for Zurich (47.38° N, 8.54° E) at 22:00 UTC on 28 August 2026. The model places 205 catalogue stars above the rendering threshold and Saturn in the visible sky.

angle H. Apparent altitude is then adjusted with Bennett's refraction formula at the default pressure and temperature used by the code [4].

The star catalogue contains 457 entries with J2000 right ascension, declination, and visual magnitude. Each entry receives a simple linear precession correction,

$$\Delta\alpha = mT + nT \sin\alpha \tan\delta, \tag{8}$$

$$\Delta\delta = nT \cos\alpha, \tag{9}$$

with the constants stored in the source. Proper motion is not applied.

3 Seal format

The sky is projected from altitude a and azimuth A onto a unit disc by the zenith-centred stereographic map

$$r = \tan\left(\frac{\pi/2 - a}{2}\right), \qquad x = r\sin A, \qquad y = -r\cos A. \tag{10}$$

The horizon is $r = 1$. Stars are drawn as black circles with a small magnitude hierarchy. The horizon, an altitude guide, a meridian, cardinal marks, the ecliptic, the Moon, and visible planets provide the remaining structure. The planetary marks use conventional astronomical symbols. The SVG has no visible text; the digest is returned separately by the Python API.

The digest is an eight-byte BLAKE2b digest encoded with a Base58 alphabet. It is a compact identifier for the supplied values and selected lunar fields. It is not a reversible encoding, a signature, or a guarantee that two different inputs cannot collide.

4 Worked reconstruction

The following example tests the inverse routine under controlled conditions. A seal is generated for Zurich at 22:00 UTC on 28 August 2026, with elevation 400 m. The solver receives the projected coordinates together with their catalogue or planet names, the same elevation, and an initial guess of $(48°, 7.5°, JD - 0.01)$. It minimizes

$$\sum_i \left\| \mathbf{p}_i(\phi, \lambda, JD) - \mathbf{y}_i \right\|^2 \tag{11}$$

by finite-difference Gauss–Newton steps with damping and backtracking.

The fit returns $\phi = 47.376899995°$, $\lambda = 8.541696°$, and $JD = 2461281.416666677$ after 14 iterations. Its RMS projected residual is 2.5×10^{-10} in model units. This is a consistency result: the observations and the fit use the same approximate forward model. It does not show that an unlabeled printed seal can identify its stars, its host planet, or its date without additional information. Those problems are outside this implementation.

5 Requirements for a broader timestamp

A single sky drawing is not enough to identify a host planet or an absolute place and time. A useful civilization-agnostic record would need more than the current seal.

First, the record would need a geometric calibration: the horizon and zenith, handedness, angular scale, orientation marks, and a machine-readable legend. Alchemical symbols are useful visual marks, but they are cultural conventions; a universal decoder should use distinct geometric marks or carry its legend in the same construction.

Second, it would need observations at several times, with the intervals stated in a physical time scale. A snapshot gives positions, while motion identifies the moving bodies. A dynamical fit over candidate host bodies could then distinguish a planet, moon, or spacecraft. The fit would need a stated search interval because celestial configurations recur.

Third, it would need an explicit reference frame and model description: the time scale (for example TT or TDB), the ephemeris version and coefficients, the star catalogue, the observer height, the reference pole, and the definition of longitude. There is no convention-free absolute longitude; longitude is always relative to a chosen meridian.

Finally, the record should contain the measured geometry and uncertainties, with redundancy and error correction. A digest alone is one-way: it can check supplied data, but it cannot reconstruct them and cannot establish that a clock was correct. These additions define a possible protocol; they are not implemented by the present script.

6 Limits and conclusion

The repository includes deterministic checks for Julian-date conversion, the Kepler residual, lunar distance scaling, stereographic round trips, input validation, and the controlled inverse fit. These checks establish implementation consistency, not absolute astronomical accuracy. The model is intended for compact drawings and reproducible software experiments, not navigation, surveying, historical dating, or precision ephemeris work. Its main limits are the low-order planetary elements, selected lunar terms, simple star correction, ERA-based rotation proxy, fixed refraction conditions, and the absence of ΔT, UT1, nutation, aberration, light-time, and proper motion. Starhash makes one small choice: use a portable analytical sky model to draw a quiet, label-free seal and return a digest beside it. The value is in that format, not in competing with astronomical libraries.

References

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